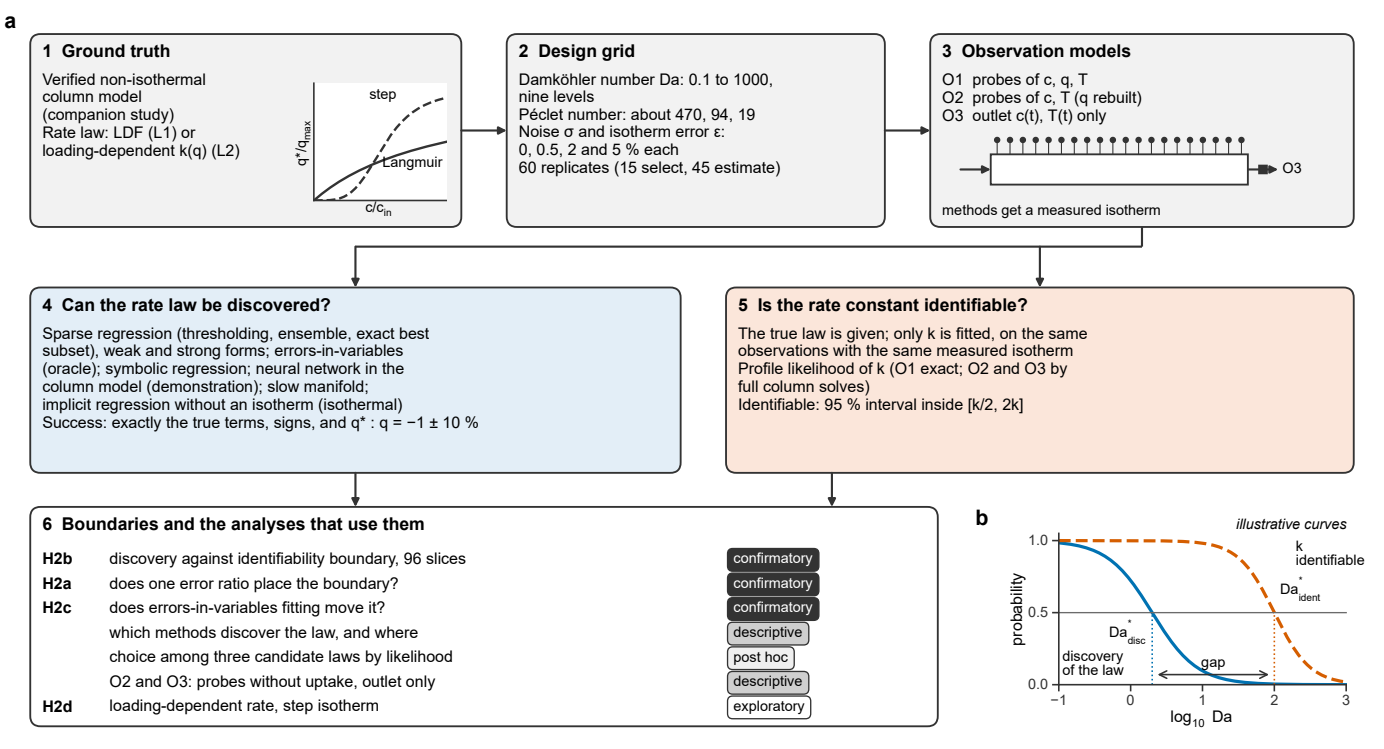

a
1 Ground truth
Verified non-isothermal column model (companion study)
Rate law: LDF (L1) or loading-dependent k(q) (L2)
step
Langmuir
q*/q_max
c/c_in
2 Design grid
Damköhler number Da: 0.1 to 1000, nine levels
Péclet number: about 470, 94, 19
Noise σ and isotherm error ε: 0, 0.5, 2 and 5 % each
60 replicates (15 select, 45 estimate)
3 Observation models
O1 probes of c, q, T
O2 probes of c, T (q rebuilt)
O3 outlet c(t), T(t) only
O3
methods get a measured isotherm
4 Can the rate law be discovered?
Sparse regression (thresholding, ensemble, exact best subset), weak and strong forms; errors-in-variables (oracle); symbolic regression; neural network in the column model (demonstration); slow manifold; implicit regression without an isotherm (isothermal)
Success: exactly the true terms, signs, and q* : q = −1 ± 10 %
5 Is the rate constant identifiable?
The true law is given; only k is fitted, on the same observations with the same measured isotherm
Profile likelihood of k (O1 exact; O2 and O3 by full column solves)
Identifiable: 95 % interval inside [k/2, 2k]
6 Boundaries and the analyses that use them
H2b discovery against identifiability boundary, 96 slices confirmatory
H2a does one error ratio place the boundary? confirmatory
H2c does errors-in-variables fitting move it? confirmatory
which methods discover the law, and where descriptive
choice among three candidate laws by likelihood post hoc
O2 and O3: probes without uptake, outlet only descriptive
H2d loading-dependent rate, step isotherm exploratory
b
illustrative curves
k identifiable
Da*ident
Da*disc
discovery of the law
gap
probability
1.0
0.5
0.0
−1
0
1
2
3
log10 Da